OOAD FINAL PROJECT · GROUP 1

From OOAD design to a working farm

Smart Farm Monitoring System

Eleven deliverables, each shown as concept, diagram, real Java code and the running app, plus this week’s new devices, sensor repair and thermostat rules.

github.com/RamekhCHHOENG/smart-farm-monitoring

HOW EVERY CHAPTER WORKS

Each deliverable is shown four ways

1

Concept

What the deliverable means, in plain words.

2

Diagram

The UML from our design, rendered from the PlantUML sources in the repo.

3

Real code

The Java that implements it, with its file and line numbers.

4

Running app

The screen where you can see it work.

New this week: 3 sensor kinds and 4 actuator kinds, a repair flow for sensors, thermostat rules with a fan-heater interlock, and a Devices page.

DELIVERABLE 1 · PROBLEM STATEMENT

Farms are checked by hand, so problems are found too late

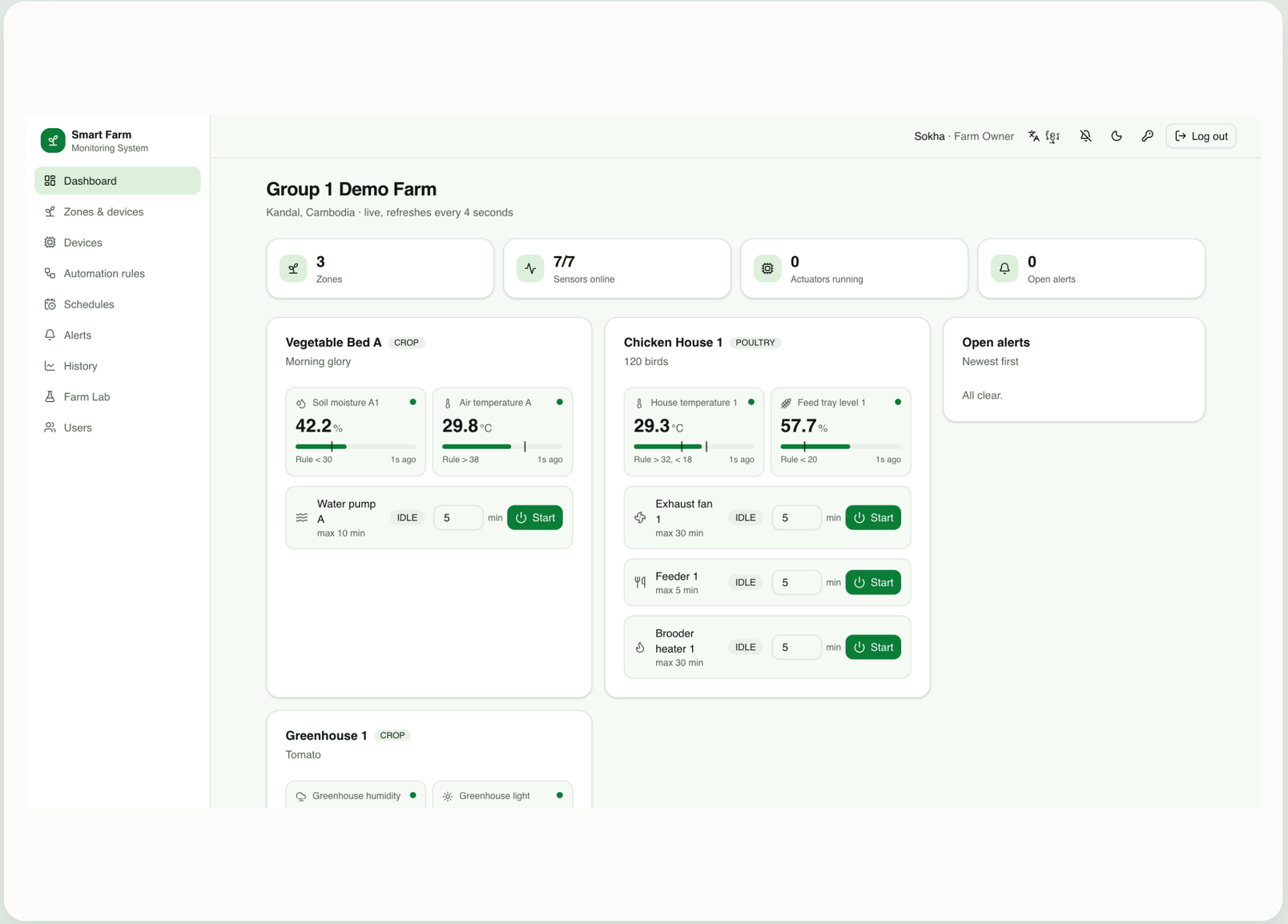
Problem	Small farms in Cambodia are checked by hand. Dry soil, heat stress in a chicken house or an empty feeder are found too late: crop loss, sick animals, wasted water and feed, long hours.
Solution	Sensors in crop beds, greenhouses and chicken houses; rules that start pumps, fans, heaters, misters and feeders by themselves; alerts to farm staff; schedules; reports.
Core value	Response in seconds, not hours · fewer losses · lower workload · decisions based on data

THE RUNNING SYSTEM

The live dashboard

A vegetable bed, a chicken house and a greenhouse, refreshing every 4 seconds.

Each tile shows the value, how close it is to its rule, and how fresh it is.



DELIVERABLE 2 · ACTORS AND PERMISSIONS

Three people and two external systems

Actor	Type	Can do
Farm Owner	Primary	Everything: zones, rules, schedules, staff accounts
Farm Worker	Primary	Watch the dashboard, start and stop devices, acknowledge alerts
Technician	Supporting	Register devices, report faults, repair actuators and sensors
Sensor Device	External system	Send readings and read its relay state, with a device key
Notification Service	External system	Deliver alerts (console or Telegram)

DELIVERABLE 2 · IN THE CODE

Permissions are checked on the server

Every write operation calls **AccessPolicy.require** with the roles allowed to do it.

The owner always passes; other roles only get what the permissions table gives them.

Hiding a button is never the only protection.

service/AccessPolicy.java

```
23 public AppUser require(Long userId, Role... allowed) {
24   if (userId == null) {
25     throw new UnauthorizedException("Please log in");
26   }
27   AppUser user = users.findById(userId)
       .orElseThrow(() -> new ForbiddenException(...));
28   if (user.getRole() == Role.OWNER
       || Arrays.asList(allowed).contains(user.getRole())) {
29     return user;
30   }
31   throw new ForbiddenException(switch (user.getRole()) {
32     case WORKER -> "A Farm Worker account cannot do this...";
33     case TECHNICIAN -> "A Technician account cannot...";
34     case OWNER -> "Not allowed";
35   });
```

DELIVERABLE 3 · FUNCTIONAL REQUIREMENTS

Each requirement is a screen and a service

ID	Requirement	Where in the app
FR-01	Farm and zones (crop, poultry)	Zones & devices · FarmService
FR-02	Register, rename, repair, remove devices	Devices · FarmService, SensorService
FR-03	Receive and validate readings	ReadingController · MonitoringService
FR-04	Live dashboard	Dashboard · QueryService
FR-05 / 06	Rules, evaluated on every reading	Automation rules · AutomationRule
FR-07	Scheduled tasks	Schedules · TaskService
FR-08	Manual control with a log	Dashboard · ActuatorService
FR-09	Alerts: breach, fault, offline	Alerts · AlertService
FR-10	History, device log, CSV	History · QueryService
FR-11	Staff accounts	Users · UserService

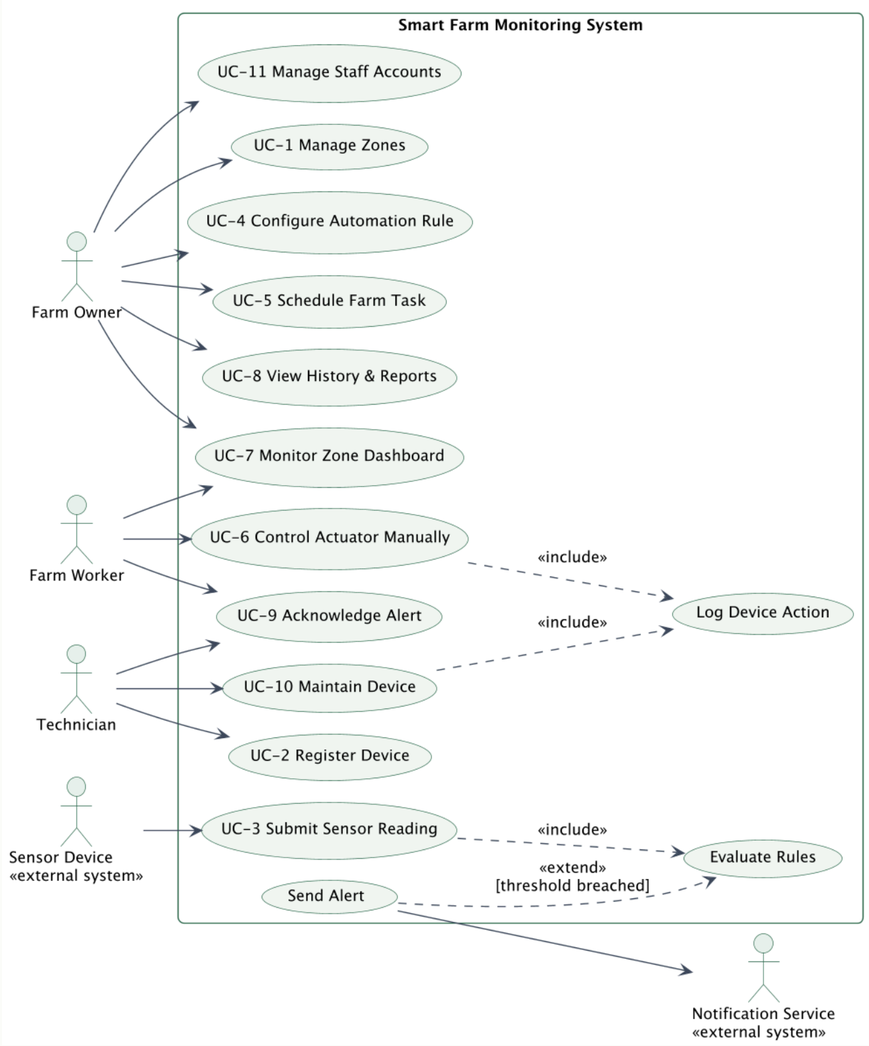
DELIVERABLE 4

Use case diagram

Eleven use cases for five actors.

«include»: every reading is evaluated against the rules; every manual action is logged.

«extend»: an alert is sent only when a threshold is breached.



DELIVERABLE 4 · IN THE CODE

Use case 3: Submit Sensor Reading

The device's request arrives at **ReadingController**, the use case's entry point.

«include» Evaluate Rules: the loop over every active rule always runs.

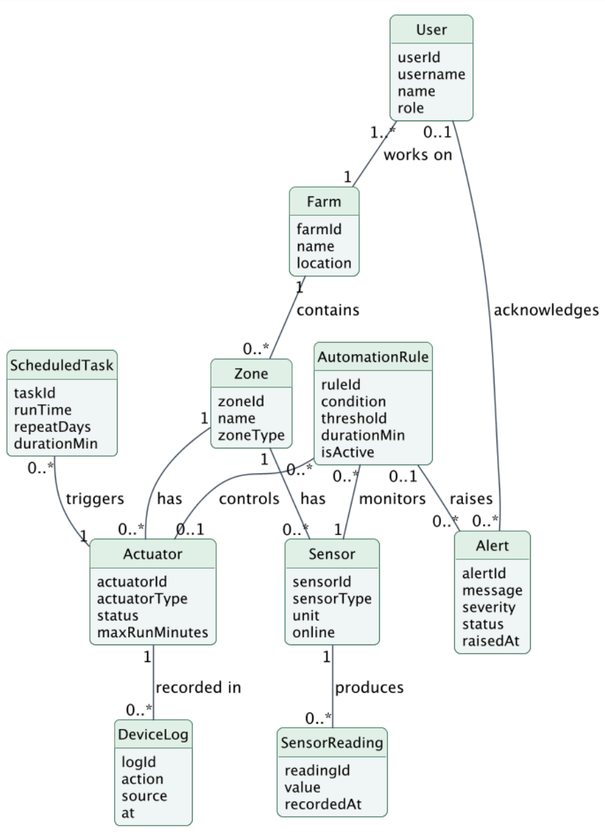
«extend» Send Alert: only inside **if (rule.isTriggered(…))**.

service/MonitoringService.java

```
49 public ReadingResult processReading(Long sensorId,
                   double value) {
51   Sensor sensor = sensors.findById(sensorId)...;
..   // faulty sensor ignored, [valid?] checked (D8)
66   SensorReading reading =
       readings.save(sensor.record(value, now));
73   List<String> actions = new ArrayList<>();
74   for (AutomationRule rule :
        rules.findBySensor_IdAndActiveTrue(sensorId)) {
75     if (rule.isTargetReached(reading)) {
77       actuatorService.stopAtTarget(rule, reading, now);
80     } else if (rule.isTriggered(reading, now)) {
81       actions.add(respond(rule, reading, now));
82     }
83   }
```

DELIVERABLE 5 · CONCEPTUAL DOMAIN MODEL

A farm contains zones; a zone owns its devices




```
domain/Zone.java
29 // Composition: Zone ◆── Sensor, Zone ◆── Actuator
30 @OneToMany(mappedBy = "zone",
       cascade = CascadeType.ALL, orphanRemoval = true)
32 private List<Sensor> sensors = new ArrayList<>();

34 @OneToMany(mappedBy = "zone",
       cascade = CascadeType.ALL, orphanRemoval = true)
36 private List<Actuator> actuators = new ArrayList<>();
```

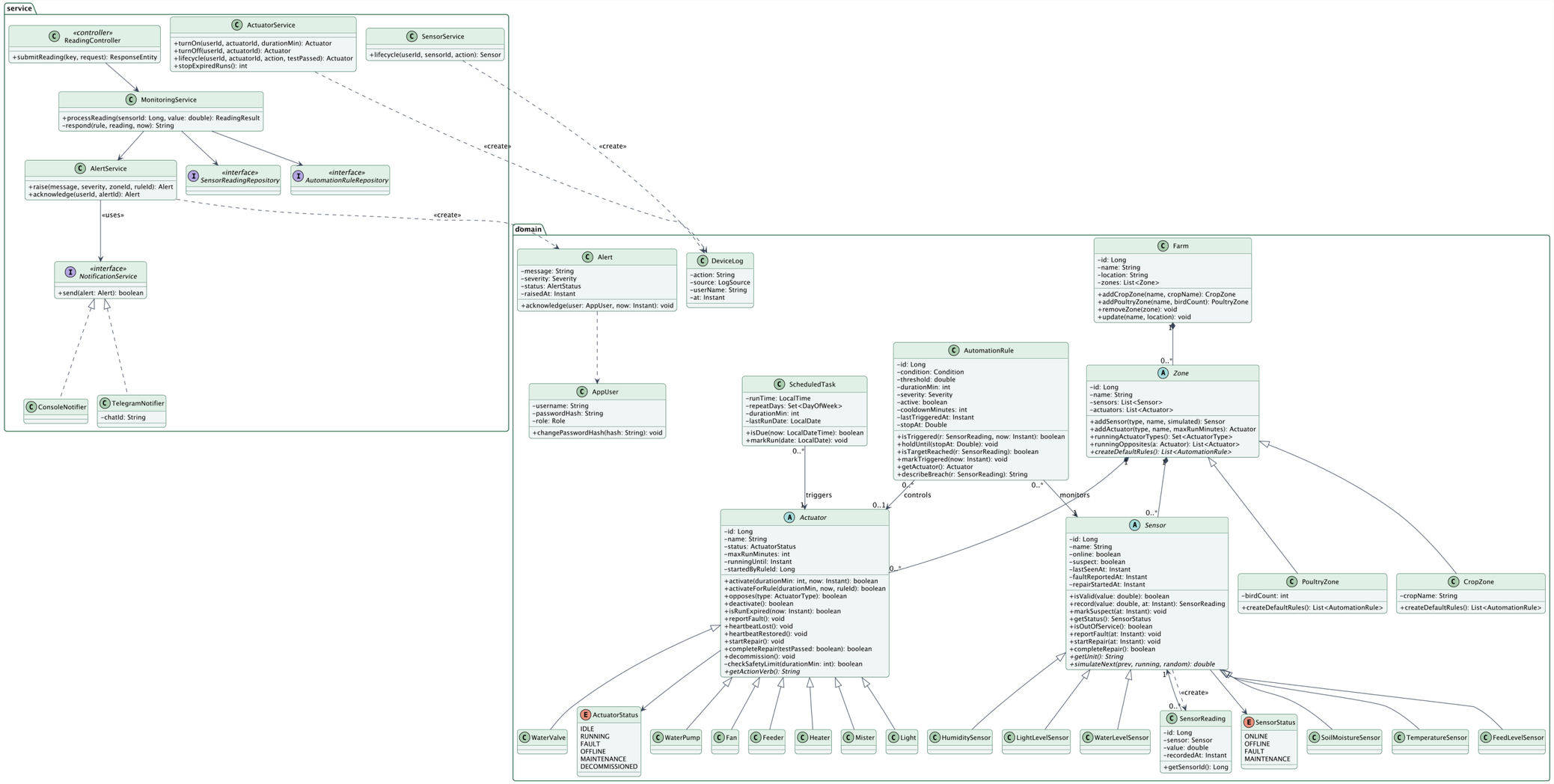
cascade = ALL and **orphanRemoval**: deleting a zone deletes its devices, readings, rules and schedules.

Multiplicity 0..* is a List that may be empty: a zone can exist before its devices are installed.

DELIVERABLE 6

Design class diagram

Association, composition, inheritance, interface realization and dependency. Now with 6 sensor and 7 actuator subclasses, SensorStatus, and the thermostat members.



OOP PILLAR 1 · ENCAPSULATION

The actuator protects its own state

status is private. Nobody can set it directly.

The only way to change it is through methods like **activate**, which check the rules first.

A pump can never be started twice, or run longer than its safety limit.

domain/Actuator.java

```
27 @Enumerated(EnumType.STRING)
28 private ActuatorStatus status = ActuatorStatus.IDLE;
..
70 public boolean activate(int durationMin, Instant now) {
71   if (status != ActuatorStatus.IDLE
         || !checkSafetyLimit(durationMin)) {
72     return false;
73   }
74   status = ActuatorStatus.RUNNING;
75   runningUntil = now.plus(Duration.ofMinutes(durationMin));
76   return true;
77 }
```

OOP PILLAR 2 · ABSTRACTION

Sensor says what every sensor must do, not how

Sensor is an **abstract class**.

Services only know these methods: type, unit, valid range, how it reacts to running actuators.

Each of the 6 concrete sensors fills in the details.

domain/Sensor.java

```
16 public abstract class Sensor {
..
143 public abstract SensorType getType();
145 public abstract String getUnit();
147 public abstract double getMinValid();
149 public abstract double getMaxValid();
151 public abstract double initialSimulatedValue();
153 /** {@code running}: actuators running in the zone */
154 public abstract double simulateNext(double previous,
        Set<ActuatorType> running, Random random);
```

OOP PILLAR 3 · INHERITANCE

A humidity sensor IS-A sensor

HumiditySensor extends Sensor and inherits recording, validation, offline detection and the repair flow.

It only adds what is specific: %, 0 to 100, rises with the mister, falls with the fan.

Added this week in one small class, like the light and tank sensors.

domain/HumiditySensor.java

```
12 public class HumiditySensor extends Sensor {
23   @Override public SensorType getType() {
       return SensorType.HUMIDITY; }
24   @Override public String getUnit() { return "%"; }
26   @Override public double getMaxValid() { return 100; }
..
30   @Override
31   public double simulateNext(double previous,
         Set<ActuatorType> running, Random random) {
32     double delta = (TYPICAL - previous) * 0.03 + ...;
33     if (running.contains(ActuatorType.MISTER)) {
34       delta += 1 + random.nextDouble() * 0.8;
36     } if (running.contains(ActuatorType.FAN)) {
37       delta -= 0.4 + random.nextDouble() * 0.3;
39     } return clamp(previous + delta);
```

OOP PILLAR 4 · POLYMORPHISM

One call, different behaviour per zone

zone.createDefaultRules() is one call.

A **CropZone** answers with water, mist and tank rules; a **PoultryZone** with fan, heater and feed rules.

The service never asks which kind of zone it has.

domain/PoultryZone.java

```
27 @Override
28 public List<AutomationRule> createDefaultRules() {
29   List<AutomationRule> rules = new ArrayList<>();
32   addRule(rules, "Hot house - cool " + getName(),
       TEMPERATURE, ABOVE, 32, FAN, 30, WARNING, 29.0);
34   addRule(rules, "Low feed - refill " + getName(),
       FEED_LEVEL, BELOW, 20, FEEDER, 5, WARNING, 80.0);
36   addActuatorRule(rules, "Cold house - heat " + ...,
       TEMPERATURE, BELOW, 18, HEATER, 30, WARNING, 21.0);
38   addActuatorRule(rules, "Damp house - ventilate " ...,
       HUMIDITY, ABOVE, 85, FAN, 30, WARNING, 75.0);
..   // dim house: light; low tank: valve
44   return rules;
45 }
```

NEW THIS WEEK · DEVICE KINDS

6 sensors and 7 actuators, each one small class

Sensor	Unit	Usually drives
Soil moisture	%	Water pump
Temperature	°C	Fan or heater
Feed level	%	Feeder
Air humidity · new	%	Mister or fan
Light level · new	lux	Light
Water tank · new	%	Tank valve

Actuator	Limit	Action
Water pump	10 min	Watering
Fan	30 min	Cooling
Feeder	5 min	Feeding
Heater · new	30 min	Heating
Mister · new	5 min	Misting
Light · new	120 min	Lighting
Tank valve · new	15 min	Refilling

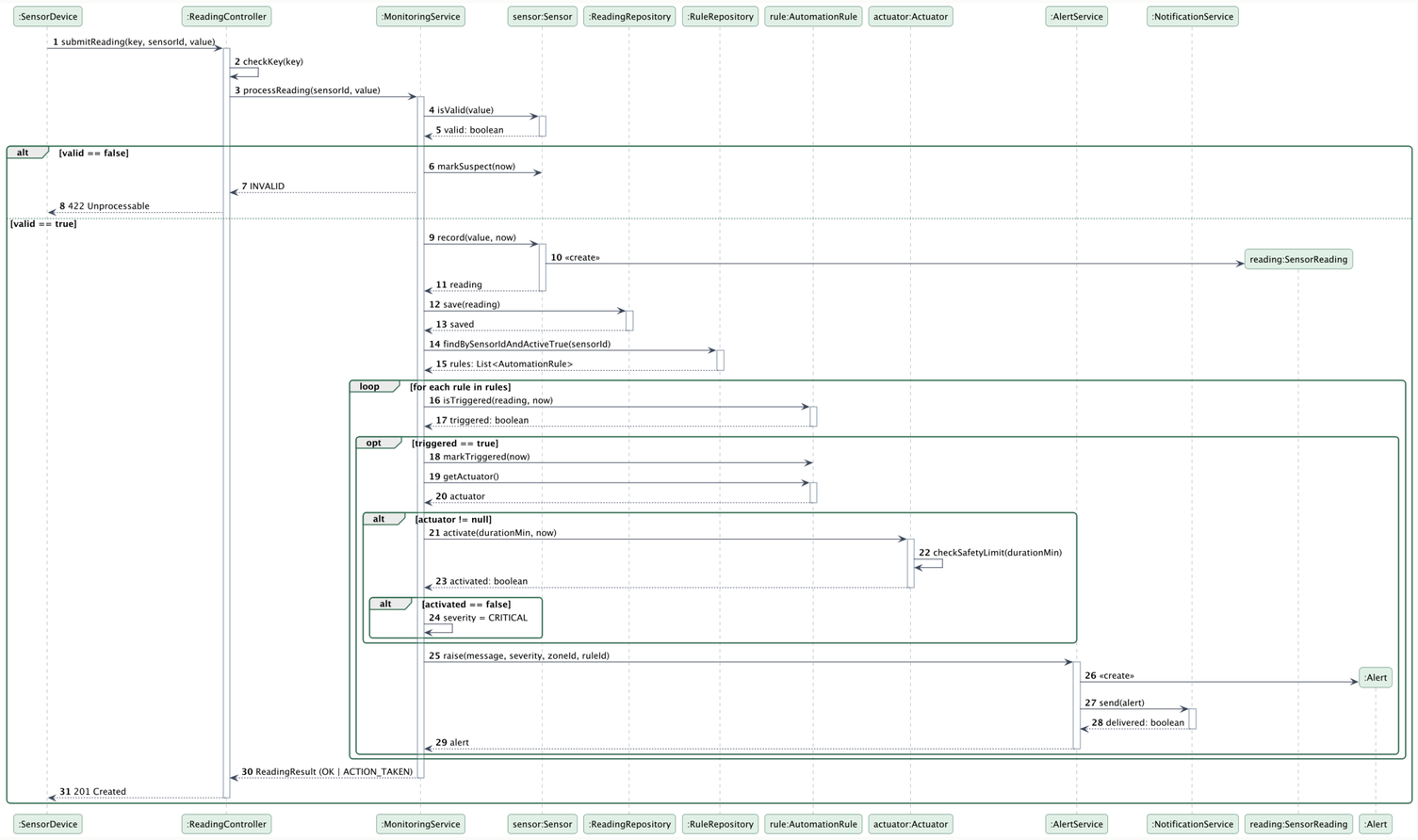
Services, screens and the Farm Lab handle a new kind with no special cases. A startup upgrade lets an older database accept them.

DELIVERABLE 7

Sequence: Submit Sensor Reading

Device → ReadingController → MonitoringService → Sensor → rules → Actuator → AlertService → NotificationService.

loop over the rules, **alt** when the actuator refuses.



DELIVERABLE 7 · IN THE CODE

Each message is a method call

rule.getActuator(), **activateForRule(…)** and **alertService.raise(…)** are the arrows of the diagram.

alt: if the actuator refuses, the alert becomes **CRITICAL**.

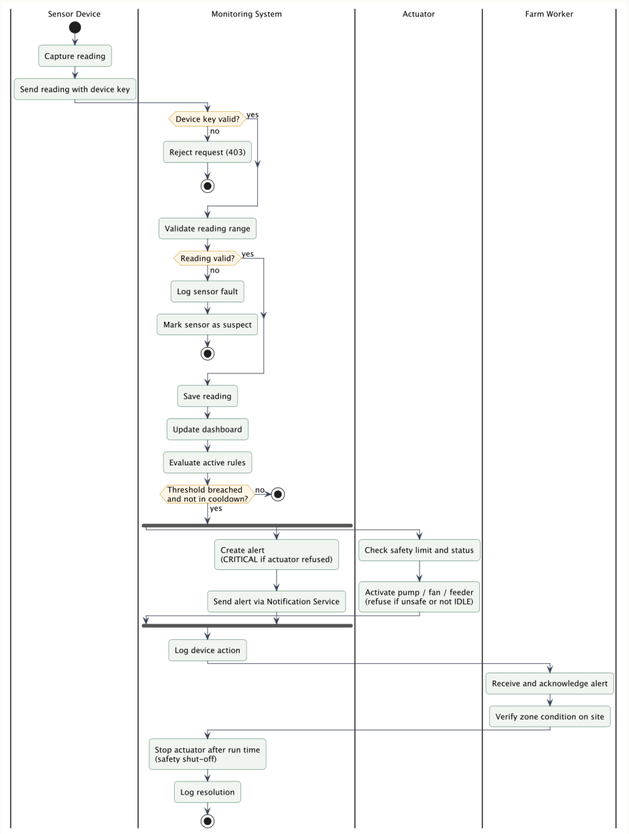
New: the interlock runs first, and a thermostat rule says where it will stop.

service/MonitoringService.java

```
90 private String respond(AutomationRule rule, ...) {
92   Actuator actuator = rule.getActuator();
96   List<String> interlocked = ... IDLE
97     ? actuatorService.stopOpposites(actuator, now) : ...;
98   if (actuator == null) { outcome = "alert only"; }
100  else if (actuator.activateForRule(
         rule.getDurationMin(), now, rule.getId())) {
102    outcome = "... started until 29°C (max 30 min)";
110  } else if (actuator.getStatus() == RUNNING) { ... }
112  else {  // refused: fault, offline, safety limit
116    severity = Severity.CRITICAL;
117  }
119  alertService.raise(rule.describeBreach(reading)
         + " - " + outcome, severity, ...);
```

DELIVERABLE 8 · ACTIVITY DIAGRAM

The [valid?] decision protects the farm




```
service/MonitoringService.java
54 if (sensor.getStatus() == SensorStatus.FAULT) {
55   return ReadingResult.ignored(...); // 202
56 }
59 if (!sensor.isValid(value)) {
60   sensor.markSuspect(now);
62   return ReadingResult.invalid(...); // 422
64 }
66 readings.save(sensor.record(value, now));
69 if (sensor.isOutOfService()) return ok; // no rule
```

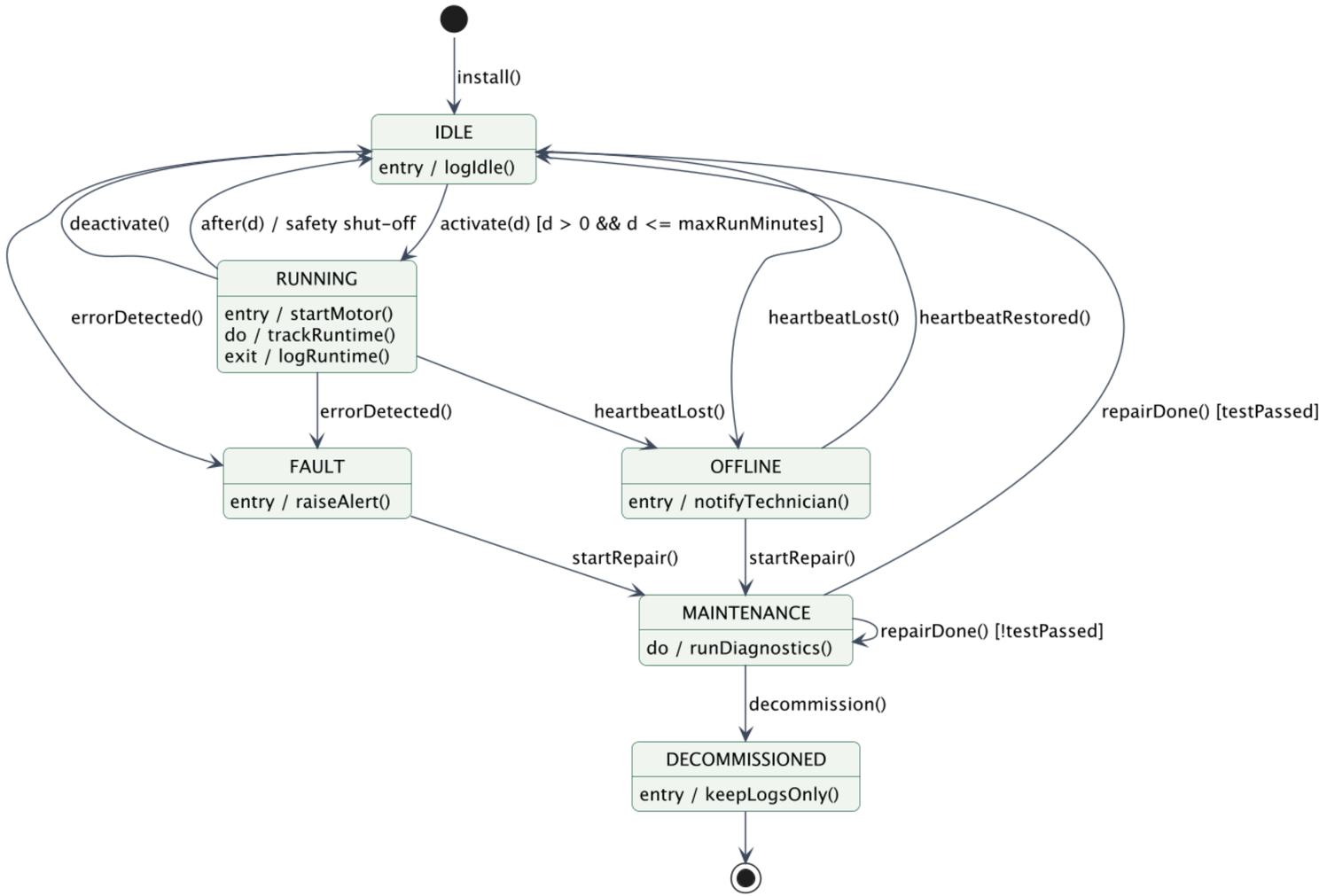
No: mark the sensor suspect, answer HTTP 422, nothing else happens. **Yes**: record it, then the fork: the actuator acts while the alert is sent.

DELIVERABLE 9

State machine: the actuator

Six states: IDLE, RUNNING, FAULT, OFFLINE, MAINTENANCE, DECOMMISSIONED.

Each transition is a method that first checks the current state. A wrong transition is refused with HTTP 409.



DELIVERABLE 9 · IN THE CODE

Guarded transitions

Each method lists the states it may start from, like the arrows of the diagram.

require(…) throws when the transition is not allowed; the API answers 409 Conflict.

A broken pump can never jump back to IDLE.

domain/Actuator.java

```
117 /** IDLE/RUNNING → FAULT */
118 public void reportFault() {
119   require(Set.of(IDLE, RUNNING), "report a fault");
120   status = ActuatorStatus.FAULT;
121   runningUntil = null;
123 }
125 /** IDLE/RUNNING → OFFLINE */
126 public void heartbeatLost() {
127   require(Set.of(IDLE, RUNNING), "mark offline");
128   status = ActuatorStatus.OFFLINE;
131 }
133 /** OFFLINE → IDLE */
134 public void heartbeatRestored() {
135   require(Set.of(OFFLINE), "restore heartbeat");
136   status = ActuatorStatus.IDLE;
```

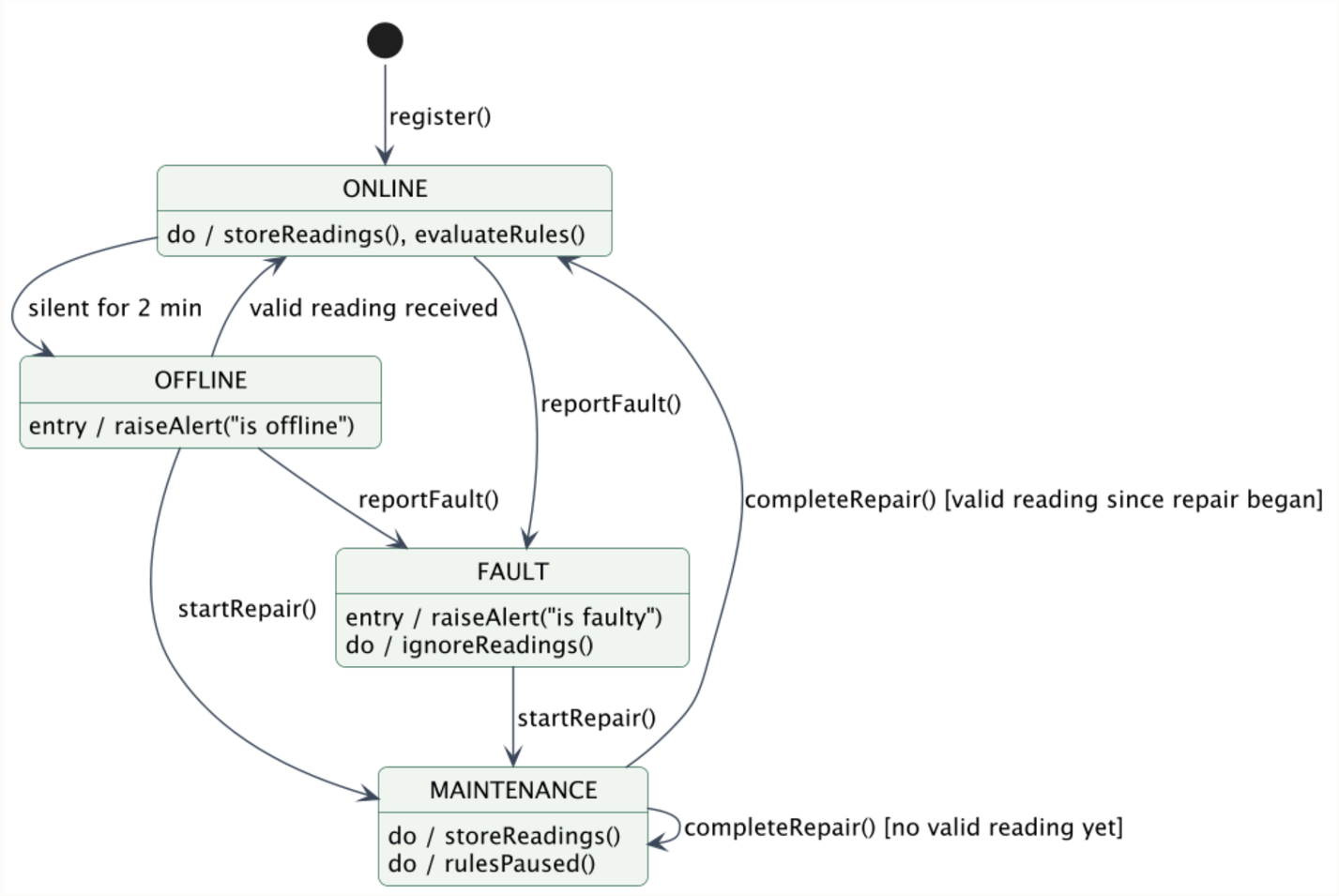
NEW · SENSOR STATE MACHINE

A broken sensor can be repaired

ONLINE ⇄ **OFFLINE** follows the signal: a sensor that stops sending comes back by itself.

FAULT: readings ignored, rules paused, so a stuck probe cannot run a pump all night.

MAINTENANCE → **ONLINE** only after a valid reading since the repair began.



NEW · IN THE CODE

The repair test is the sensor’s own data

Status is **worked out** from two timestamps, never stored twice.

Same guard pattern as the actuator: **require(…)**, else 409.

completeRepair passes only if a valid reading arrived after the repair began, not because someone clicked.

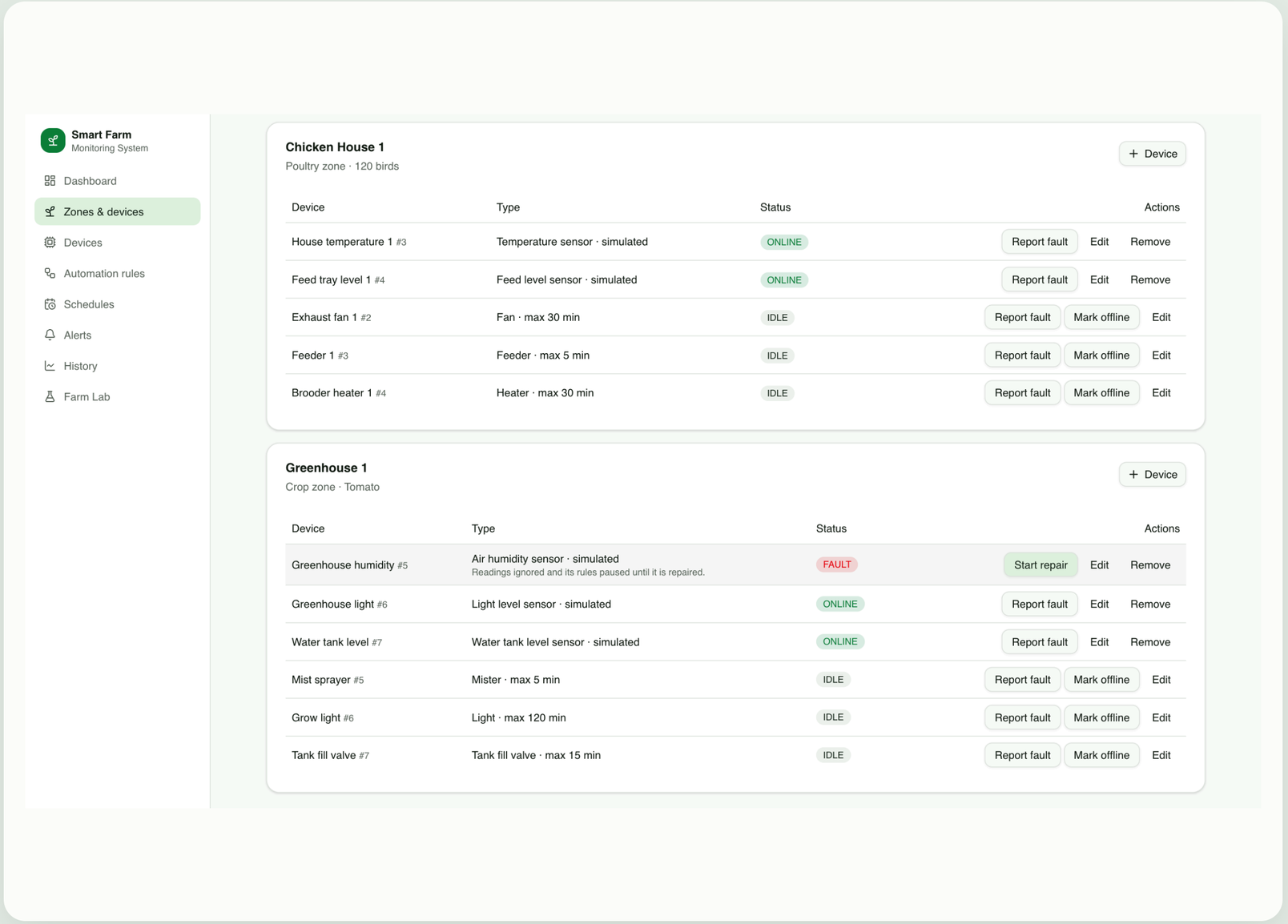
```
domain/Sensor.java
96 public SensorStatus getStatus() {
97   if (repairStartedAt != null) return MAINTENANCE;
100  if (faultReportedAt != null) return FAULT;
103  return online ? ONLINE : OFFLINE;
104 }
118 public void startRepair(Instant at) {
119  require(Set.of(FAULT, OFFLINE), "start repair");
120  repairStartedAt = at;
127 public boolean completeRepair() {
128  require(Set.of(MAINTENANCE), "complete repair");
129  boolean testPassed = online && !suspect
       && !lastSeenAt.isBefore(repairStartedAt);
130  if (testPassed) { faultReportedAt = null;
                       repairStartedAt = null; }
134  return testPassed;
```

LIVE · TECHNICIAN

Only valid actions are offered

The humidity sensor is in **FAULT**: only “Start repair” is shown, with a note saying its rules are paused.

During the repair the page refreshes itself until the test reading arrives.



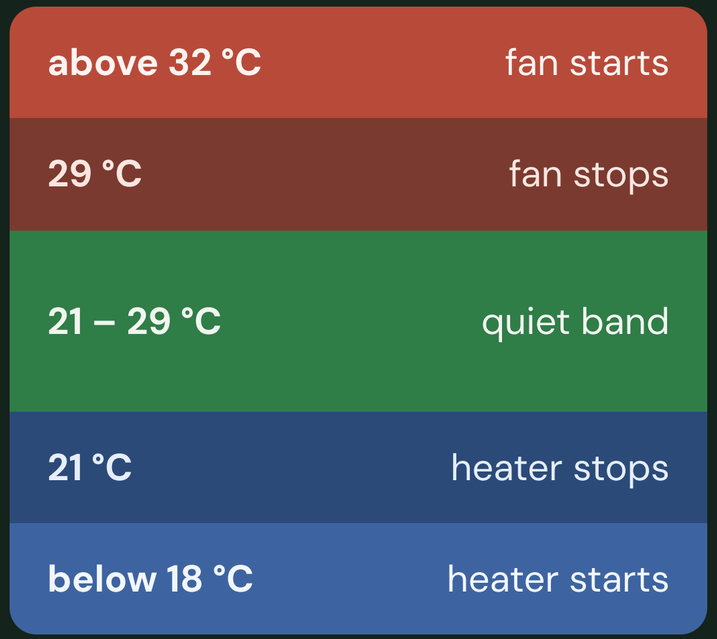
NEW · KEEPING A TEMPERATURE

A fixed 5-minute run is not enough

Before: above 34 °C the fan blows 5 minutes and stops, even if the house is still hot. Below 20 °C the heater does the same. Runs too short or too long, and fan and heater can take turns.

Now, like a thermostat: each rule has a **stop at** value. The fan runs until the house is back to 29 °C; the minutes are only a safety limit.

Interlock: fan and heater never run together. **Conflict check:** rules that would fight are refused.



NEW · HYSTERESIS IN THE CODE

The rule knows when its job is done

holdUntil refuses a stop value on the wrong side of the threshold.

isTargetReached: the reading is back in range (Information Expert again).

The actuator remembers which rule started it, so a person's run is never cut short.

```
domain/AutomationRule.java
96 public void holdUntil(Double stopAt) {
101  if (condition == ABOVE && stopAt >= threshold)
102    throw new IllegalArgumentException(
         "'Stop at' must be below " + threshold ...);
..  // and the mirror case for BELOW
110  this.stopAt = stopAt;
111 }
..
118 public boolean isTargetReached(SensorReading r) {
119  if (!keepsRange()) return false;
122  return condition == Condition.ABOVE
        ? r.getValue() <= stopAt
        : r.getValue() >= stopAt;
123 }
```

NEW · INTERLOCK

Fan and heater never run together

domain/Heater.java

```
9  public class Heater extends Actuator {
21   @Override public String getActionVerb() {
       return "Heating"; }
22   @Override
     public boolean opposes(ActuatorType o) {
       return o == ActuatorType.FAN; }
```

The heater knows its opposite; the fan says the same about the heater. Other actuators oppose nothing.

service/ActuatorService.java

```
88 List<String> stopOpposites(Actuator a, ...) {
90  for (Actuator other :
        a.getZone().runningOpposites(a)) {
91   if (other.deactivate()) {
92    deviceLogs.save(new DeviceLog(other,
93      "interlock: stopped so ... can run"));
96  }
```

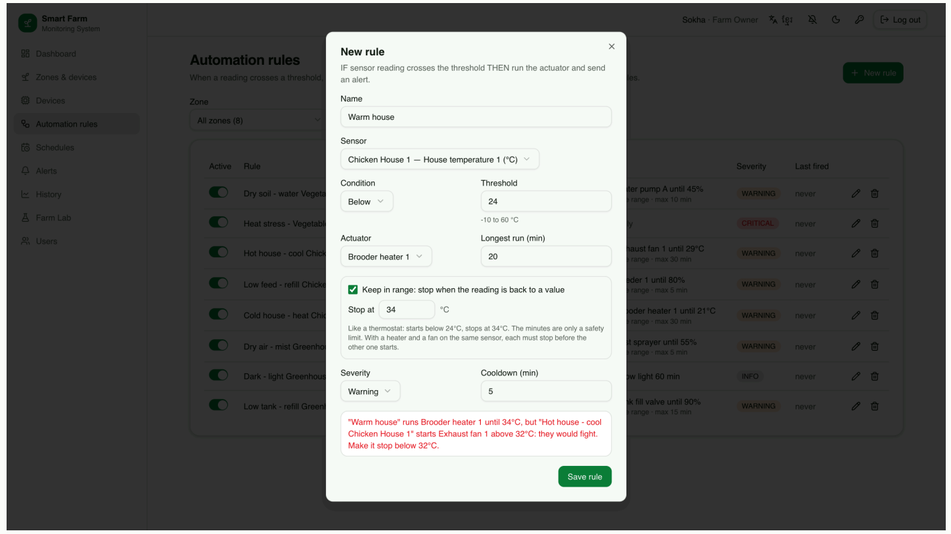
Called before a rule, a schedule or a worker starts an actuator. The zone’s log says why the other one stopped.

NEW · CONFLICT CHECK

Rules that would fight are refused

```
service/RuleService.java
83 private void checkNoFight(AutomationRule rule) {
87  for (AutomationRule other : rules on this sensor) {
88   // skip itself, alert-only rules, same direction
92   upper = the ABOVE rule; lower = the BELOW rule;
95   if (lower.getStopValue() >= upper.getThreshold())
96    throw ... "they would fight";
101  if (upper.getStopValue() <= lower.getThreshold())
102   throw ... "they would fight";
```

Each rule must stop before the other one starts, leaving a quiet band between them. The message names both rules and what to change.

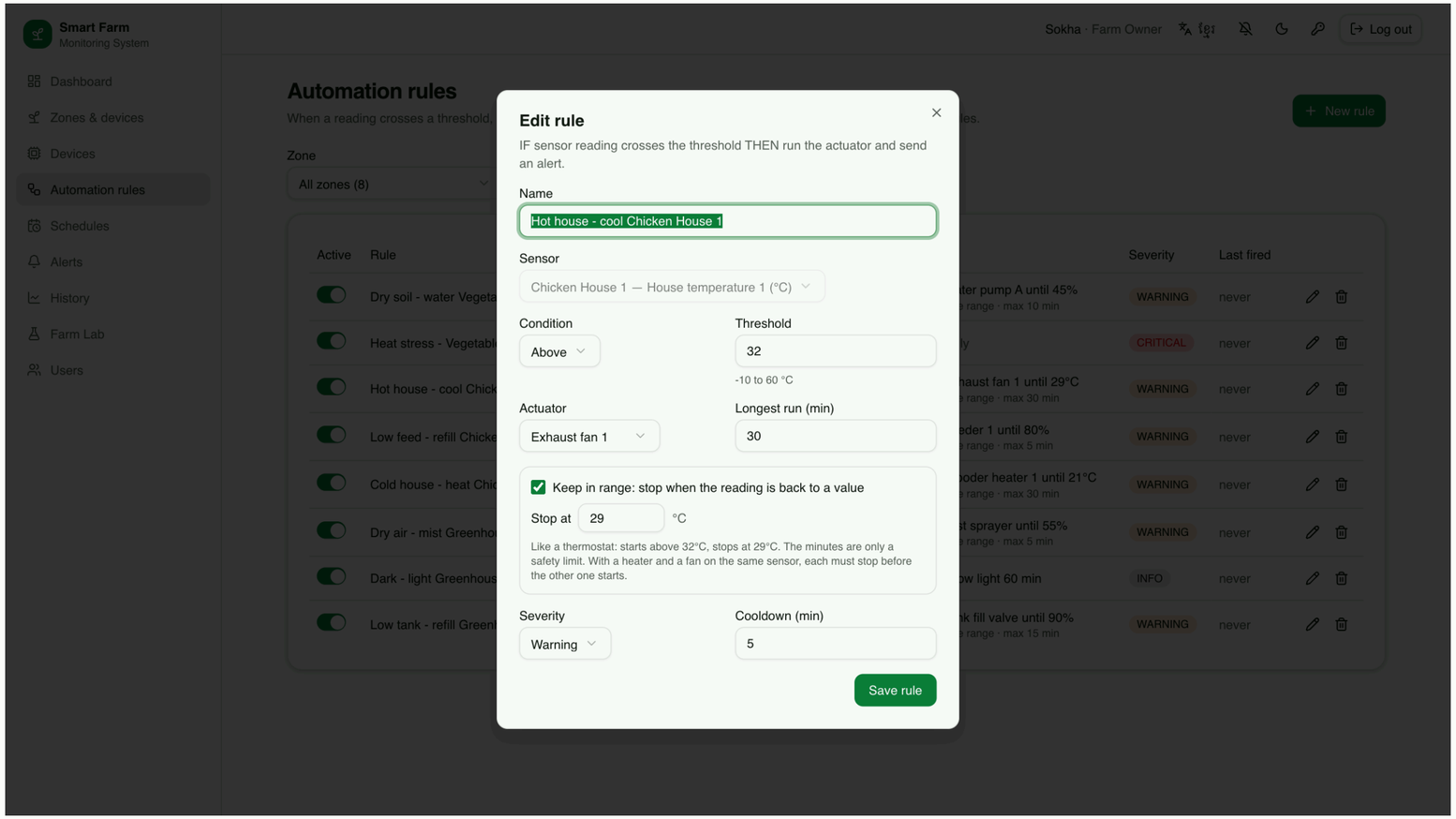


LIVE · AUTOMATION RULES

Keep in range, in the form

Tick **Keep in range**, set **Stop at**; the minutes become the longest run.

The rules list can be filtered by zone and searched.



DELIVERABLE 10 · GRASP

Information Expert

Give the job to the class that has the information.

The **rule** owns condition, threshold and cooldown, so the rule decides whether a reading triggers it.

The **sensor** knows its range (**isValid**); the **heater** knows its opposite.

domain/AutomationRule.java

```
76 public boolean isTriggered(SensorReading reading,
                              Instant now) {
77   if (!active || !reading.getSensorId()
         .equals(sensor.getId())) {
78     return false;
79   }
80   boolean breached = switch (condition) {
81     case ABOVE -> reading.getValue() > threshold;
82     case BELOW -> reading.getValue() < threshold;
83   };
84   return breached && !isCoolingDown(now);
85 }
```

DELIVERABLE 10 · GRASP

Creator

A class creates the objects it contains or records.

Farm creates its zones, **Zone** creates its sensors and actuators, and **Sensor.record** creates the **SensorReading**, after validating the value.

domain/Sensor.java

```
68 /** GRASP Creator: a sensor produces readings,
       so it creates SensorReading. */
69 public SensorReading record(double value, Instant at) {
70   if (!isValid(value)) {
71     throw new IllegalArgumentException(
72       "Value " + value + " is outside the valid range");
73   }
74   lastSeenAt = at;
75   online = true;
76   suspect = false;
77   return new SensorReading(this, value, at);
78 }
```

DELIVERABLE 10 · GRASP

Controller, Low Coupling, High Cohesion

Controller: ReadingController receives the device's system event and delegates; it holds no farm logic.

Low coupling: the device knows one endpoint.

High cohesion: each service has one focused job.

web/ReadingController.java

```
33 /** UC-3: 201 stored, 202 faulty sensor, 422 invalid */
34 @PostMapping("/readings")
35 public ResponseEntity<ReadingResult> submitReading(
       @RequestHeader("X-Device-Key") String key,
36     @Valid @RequestBody ReadingRequest request) {
37   checkKey(key);
38   ReadingResult result = monitoringService
       .processReading(request.sensorId(), request.value());
39   HttpStatus status = result.hasInvalidValue()
       ? UNPROCESSABLE_CONTENT
40     : result.isIgnored() ? ACCEPTED : CREATED;
41   return ResponseEntity.status(status).body(result);
```

DELIVERABLE 11 · SOLID

Single Responsibility: one reason to change

Class	Its one job
AutomationRule	Decide whether a reading breaks the rule, or is back in range
Actuator	Act, and guard its own state
Sensor	Validate, record, and guard its repair state
MonitoringService	Run the workflow for one reading
ActuatorService · SensorService	Manual control, interlock, technician actions
RuleService	Save rules that are valid and do not fight
AlertService · NotificationService	Create alerts · deliver a message
SchemaUpgrade	Let an older database accept new device kinds

DELIVERABLE 11 · SOLID

Open/Closed: we just did it

Open for extension: this week the heater, mister, light, tank valve and three sensors were added as new subclasses.

Closed for modification: MonitoringService, RuleService and the screens did not need a special case.

Honest limit: the small **create** factory still needs one new line per kind.

```
domain/Heater.java (new)
7  @Entity
8  @DiscriminatorValue("HEATER")
9  public class Heater extends Actuator {
11   static final int DEFAULT_MAX_MINUTES = 30;
16   Heater(Zone zone, String name, int maxRunMinutes) {
17     super(zone, name, maxRunMinutes);
18   }
20   @Override public ActuatorType getType() {...}
21   @Override public String getActionVerb() {...}
22   @Override public boolean opposes(...) {...}
23 }
```

DELIVERABLE 11 · SOLID

Dependency Inversion

AlertService depends on the **NotificationService** interface, never on Telegram.

ConsoleNotifier and **TelegramNotifier** implement it.

Choosing one is configuration (AppConfig), not a code change.

config/AppConfig.java

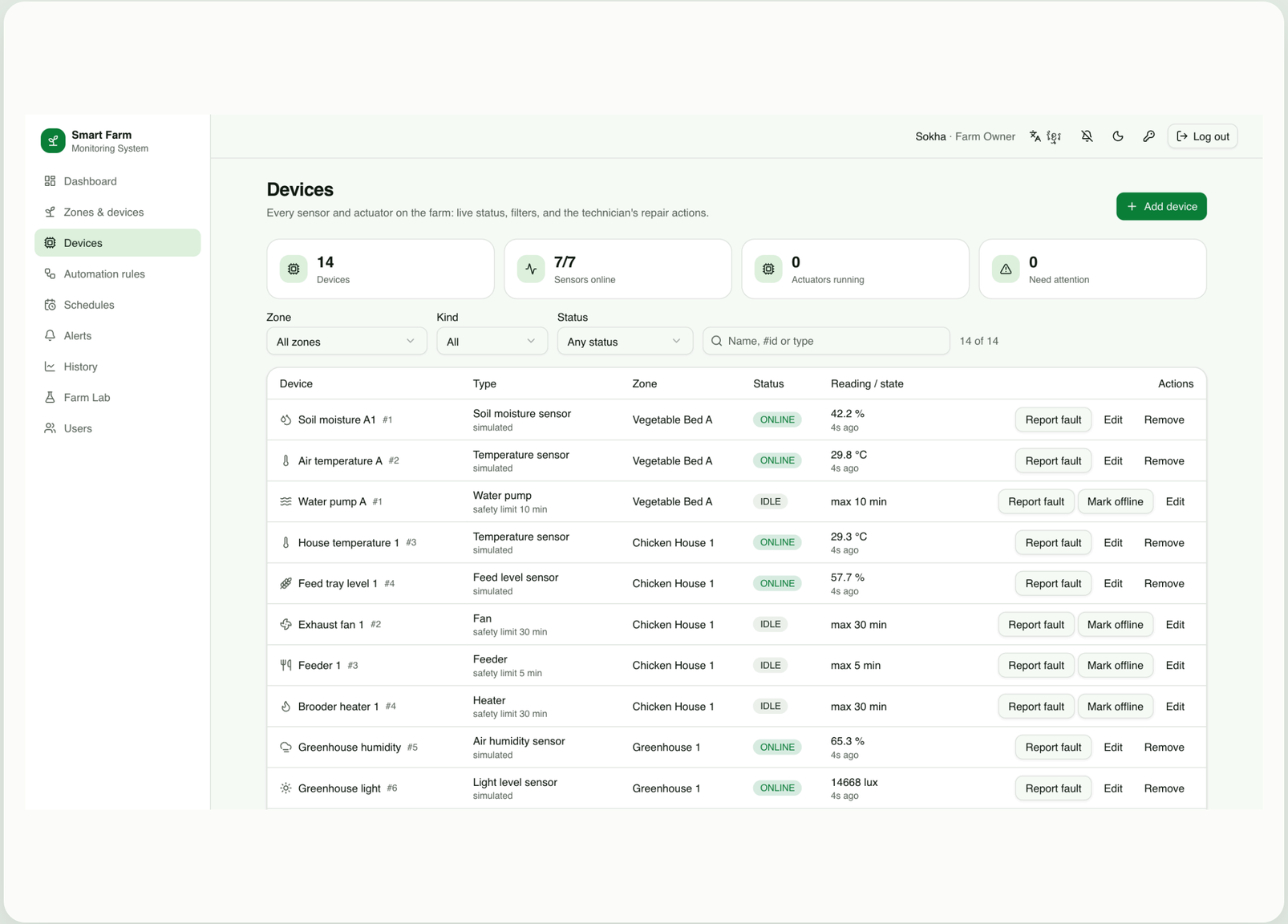
```
54 @Bean
55 public NotificationService notificationService(
       RestClient.Builder builder,
56     @Value("${farm.telegram.bot-token:}") String botToken,
57     @Value("${farm.telegram.chat-id:}") String chatId) {
58   if (!botToken.isBlank() && !chatId.isBlank()) {
59     return new TelegramNotifier(builder, botToken, chatId);
60   }
61   return new ConsoleNotifier();
62 }
```

NEW · DEVICES PAGE

Every device on one page

Live status and latest reading for every sensor and actuator.

Filter by zone, kind, status or name; “Need attention” finds faulty and offline devices.

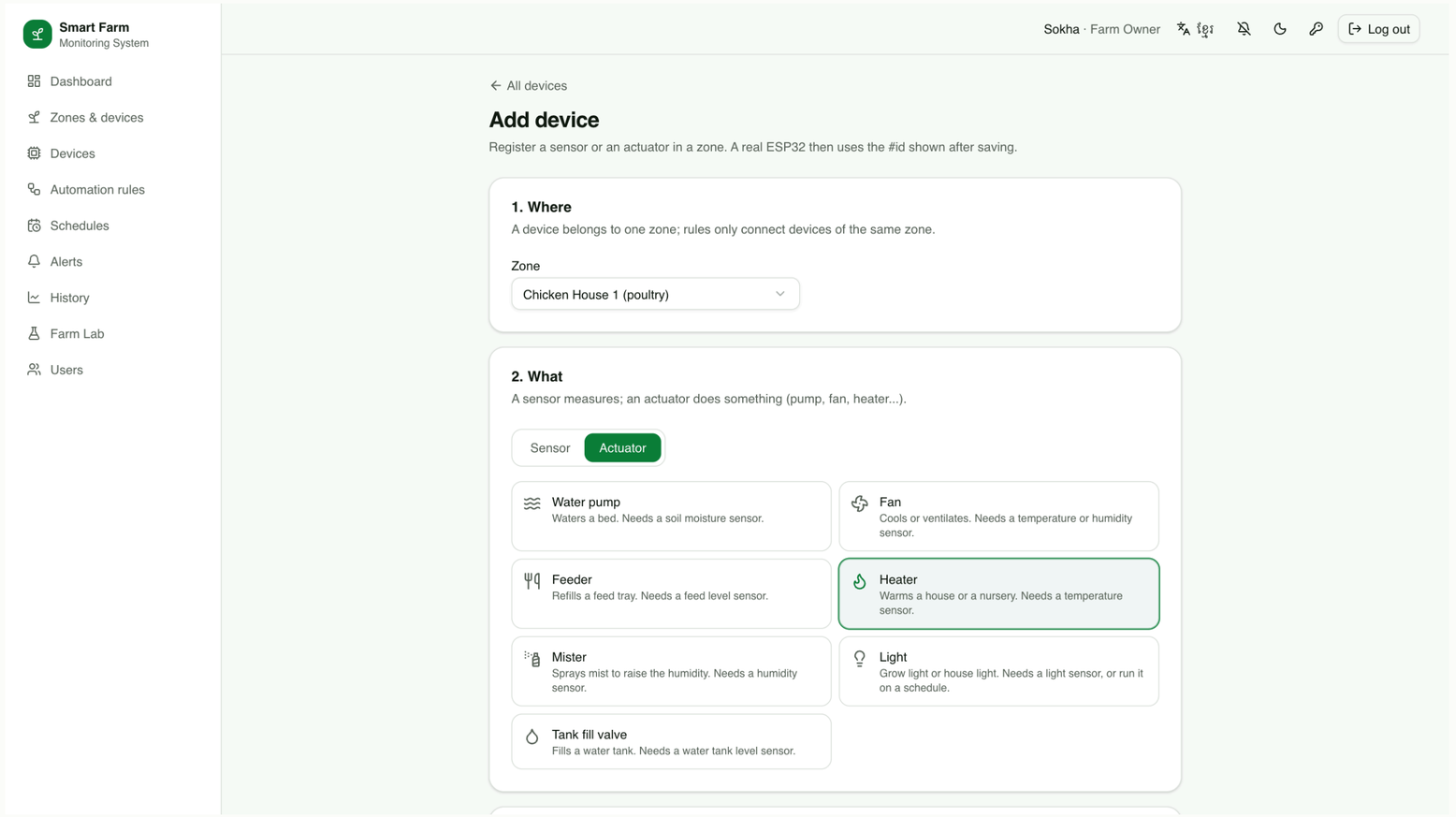


NEW · ADD DEVICE

A full page to register a device

Zone, sensor or actuator, then a type card that says what it is for.

After saving: the #id, the ESP32 line, and a link to set up a rule.

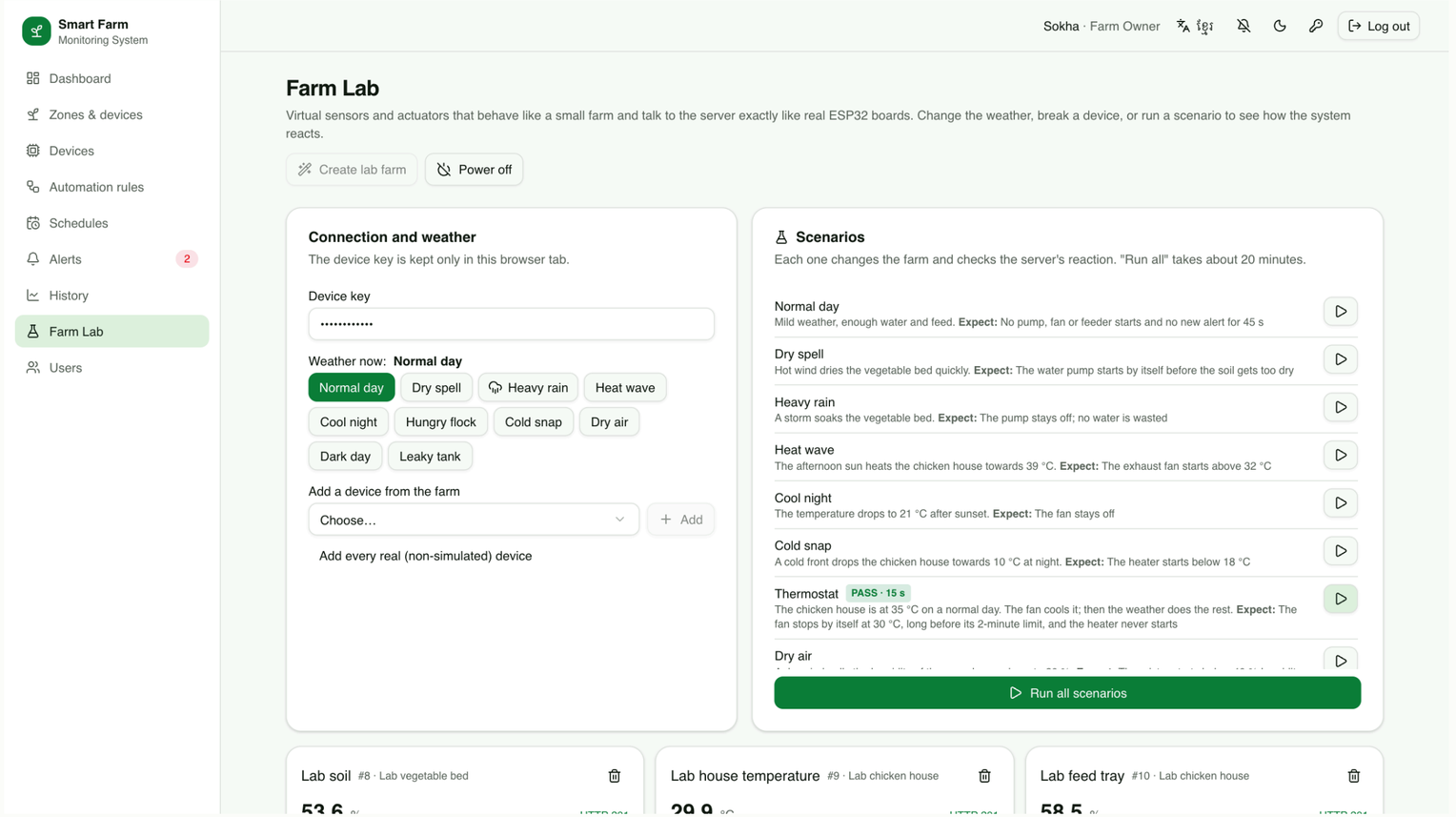


LIVE · FARM LAB

The thermostat, tested live

Virtual devices talk to the server like ESP32 boards. The house starts at 35 °C.

The fan started, then stopped itself at 30 °C after about 10 seconds, long before its 2-minute limit. The heater never started. **PASS**.



PROOF

Every rule of the design has a test

58

automated tests

Both state machines, rules and thermostat, interlock, conflict check, API flows, permissions, database upgrade.

34

browser steps

A walkthrough of every use case as owner, technician and worker, all passing.

17

farm scenarios

Drought, heat wave, cold snap, thermostat, sensor repair, power cut… One known gap is recorded: a clogged pipe.

SUMMARY

Design first, then code that follows it

Concepts → diagrams → classes and methods → tests → a farm system that works

The new features followed the same path: a state machine for sensors, hysteresis and an interlock for rules, new kinds as subclasses.

github.com/RamekhCHHOENG/smart-farm-monitoring · Thank you